

Moments and Descriptive Statistics

POLS 602
Fall 2026

Mike Burnham
Texas A&M University

Announcements

Dot Products and Norms

Dot Product

The sum of the product of corresponding numbers in two vectors.

For vectors $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$:

From components:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$$

From angle:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

Dot Product

- The dot product always returns a scalar
- Vectors must have the same number of elements to calculate
- The dot product tells you the degree to which two vectors point in the same direction

Norm

A function that assigns a strictly positive size value to a vector

L1 norm (Taxicab/Manhattan Norm): The sum of the absolute values of a vector's components.

$$\|\mathbf{a}\|_1 = \sum_{i=1}^n |a_i|$$

L2 norm (Euclidean Norm): The straight line distance.

$$\|\mathbf{a}\|_2 = \sqrt{\sum_{i=1}^n a_i^2}$$

Moments

Moments

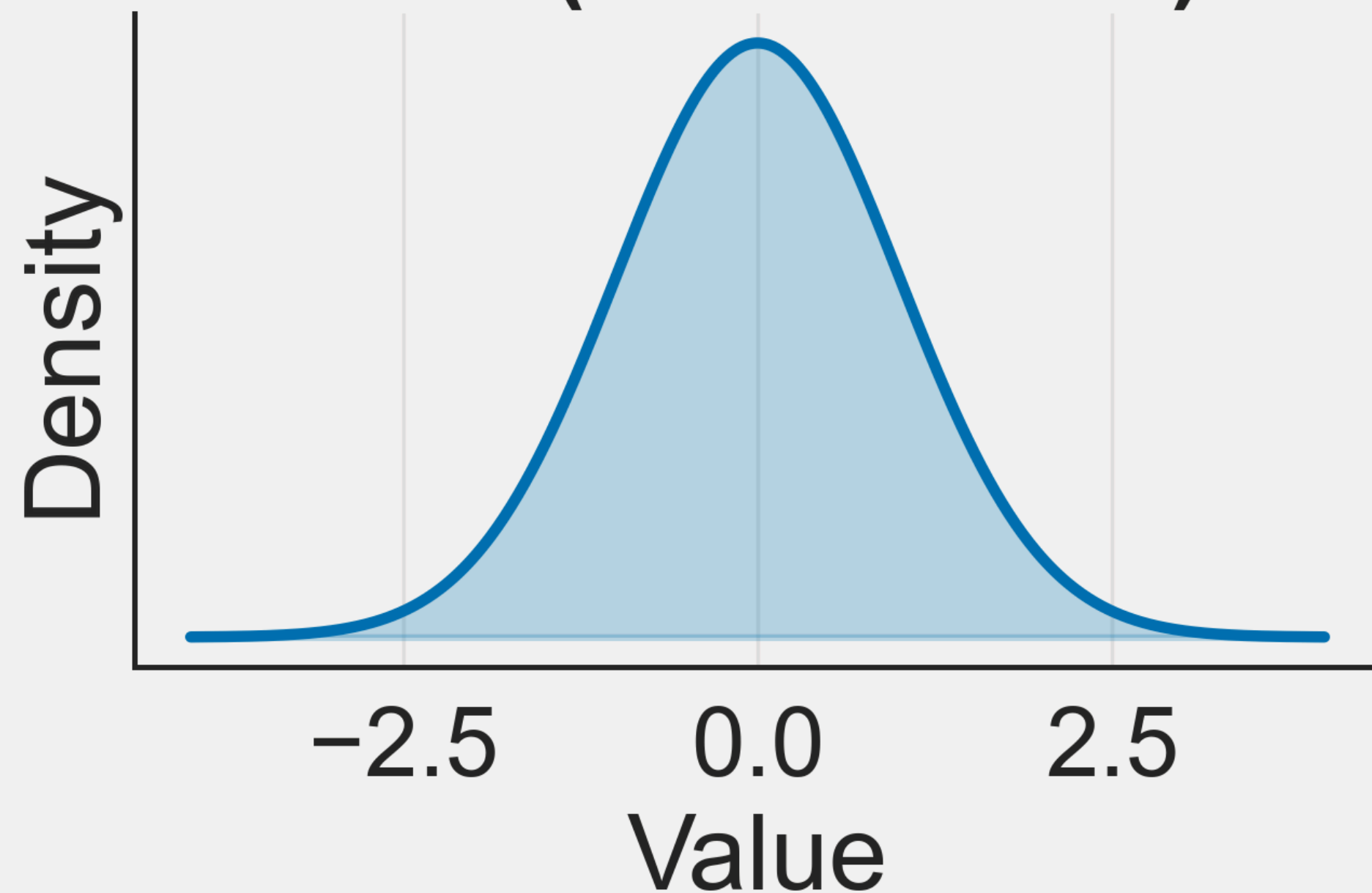
A **Moment** is *a specific quantitative measure related to the shape of a function or the properties of a distribution*

- In statistics, the function we are describing is generally a probability density function (PDF) or a probability mass function (PMF)

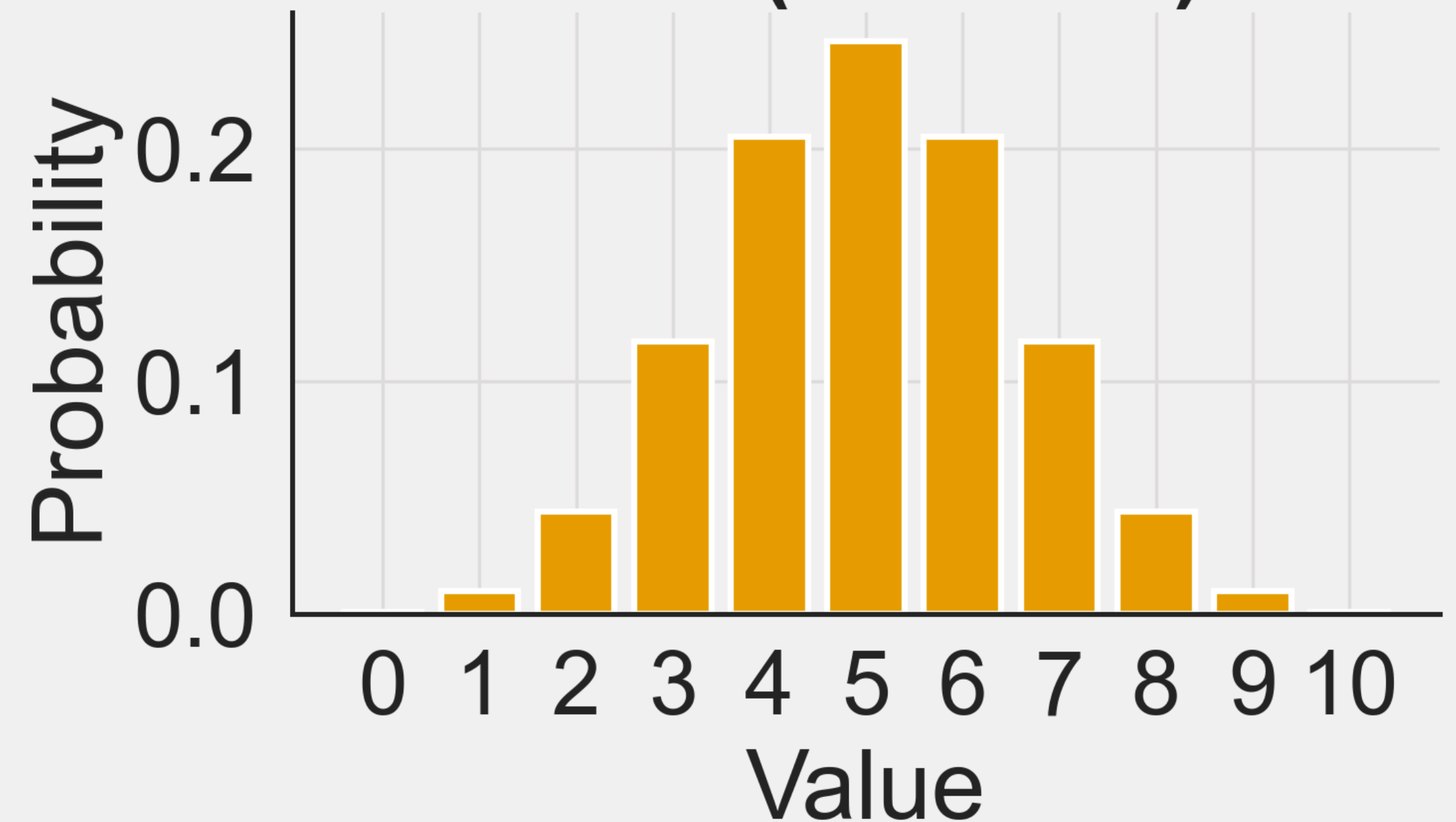
PDF vs. PMF

Continuous vs. Discrete Distributions

PDF (Continuous)



PMF (Discrete)



Moments

A **Moment** is *a specific quantitative measure related to the shape of a function or the properties of a distribution*

- In statistics, the function we are describing is generally a probability density function (PDF) or a probability mass function (PMF)
- The n -th moment of a variable X around a value of c is defined as $E[(X - c)^n]$.
- If $c = \mu$ that is the *central moment*

Central Tendency

Mean

The first moment. The sum of values divided by the number of values.

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \mathbf{1}^\top \mathbf{x}$$

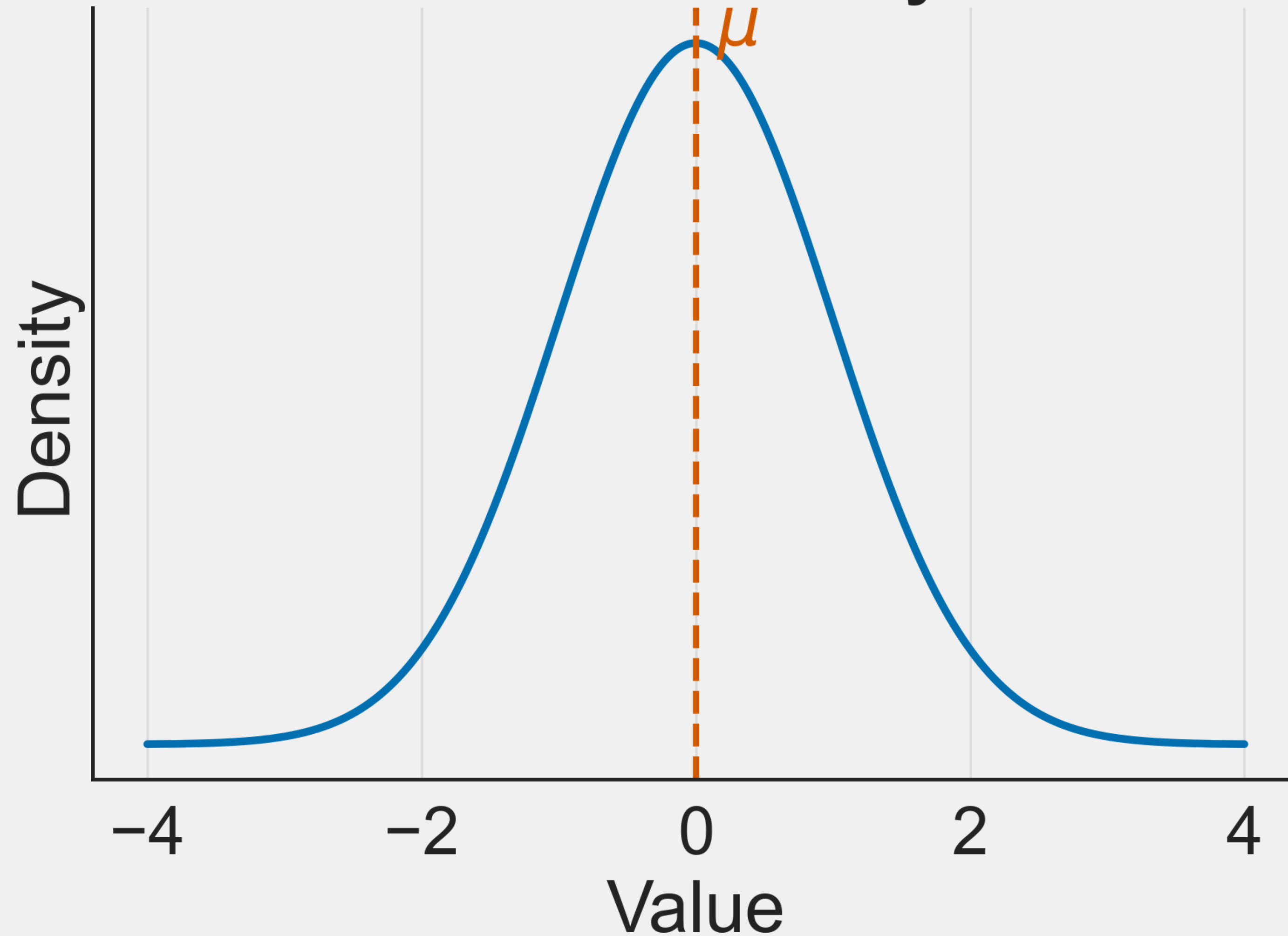
where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{1}$ is the length- n vector of ones.

Sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \mathbf{1}^\top \mathbf{x}$$

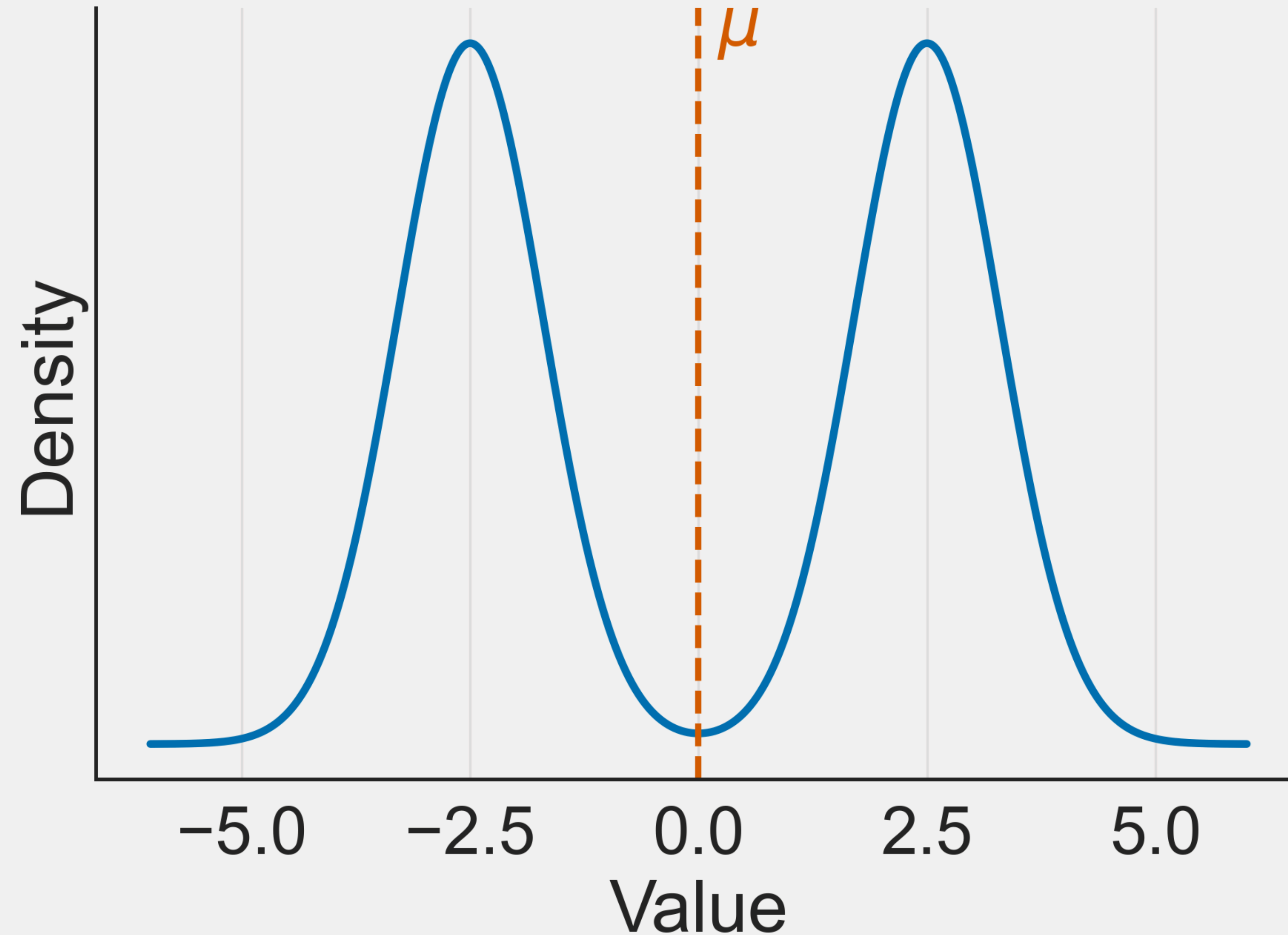
Mean

The Mean Sits at the Center of a Symmetric Distribution



Mean

The Mean Can Fall Where Data Are Least Likely



Notation

- μ — population mean (a Greek letter denotes a population *parameter*)
- $\bar{x}$ — sample mean (a Roman letter, often barred or hatted, denotes a *statistic* estimated from data)
- $\hat{\mu}$ — estimation of the population mean
- $E[Y]$ — expected value operator (used in probability theory for the theoretical mean of a random variable Y , regardless of population or sample)
- $E[Y \mid X]$ — conditional expected value operator (the expected value of Y given a specific value of X)

Median

The middle value of a sorted set of numbers.

$$\text{median}(\mathbf{x}) = \begin{cases} x_{(\frac{n+1}{2})} & n \text{ odd} \\ \frac{1}{2} \left(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)} \right) & n \text{ even} \end{cases}$$

where $x_{(i)}$ denotes the i -th smallest value of $\mathbf{x}$.

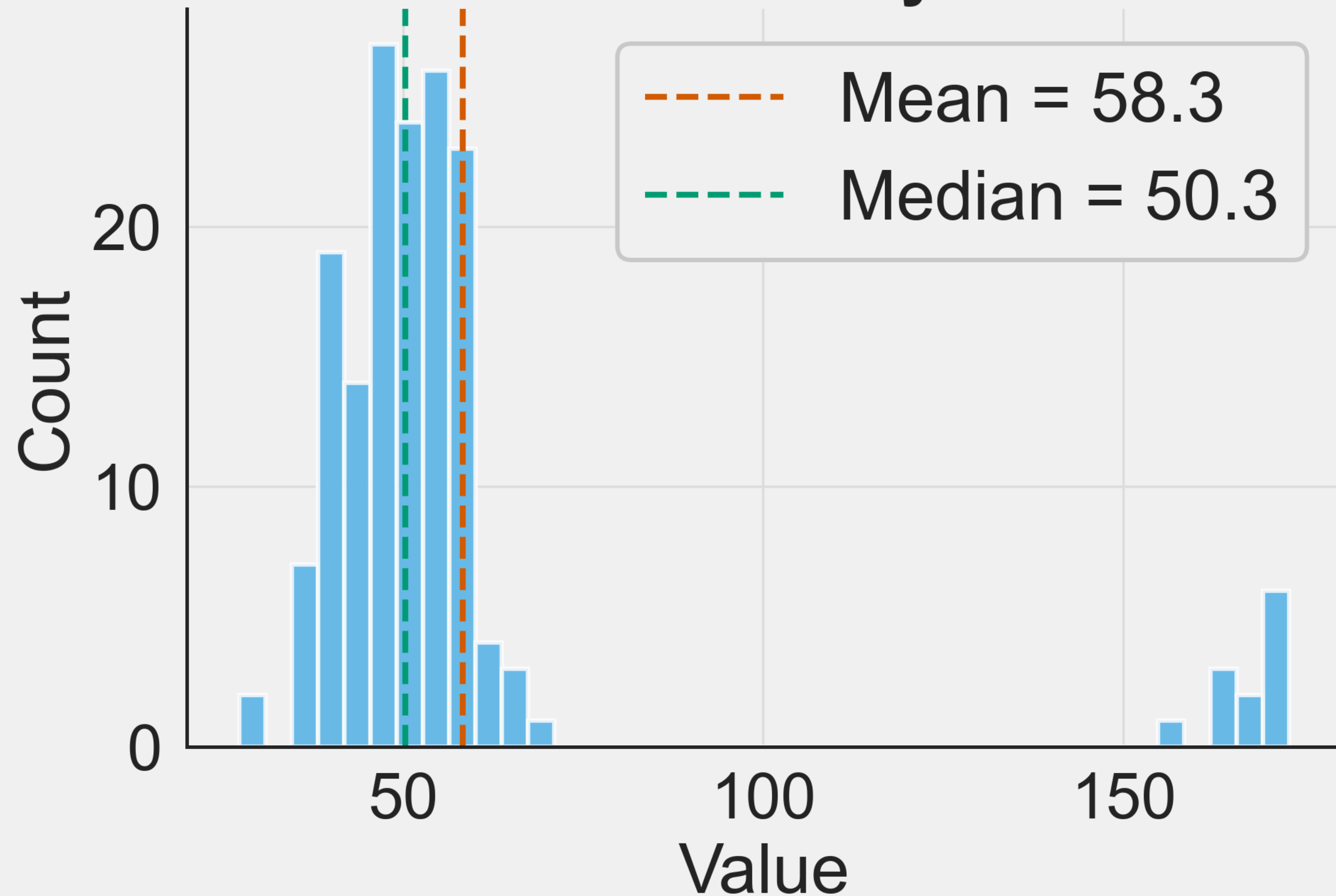
Median

The median is a **Robust** statistic.

Robust statistics are *statistics that maintain their properties even when assumptions about the underlying distribution are violated.*

Mean vs. Median

Outliers Pull the Mean Away From the Median



Spread

Variance

The second moment. The average squared deviation from the mean.

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{n} (\mathbf{x} - \mu \mathbf{1})^\top (\mathbf{x} - \mu \mathbf{1})$$

Sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} (\mathbf{x} - \bar{x} \mathbf{1})^\top (\mathbf{x} - \bar{x} \mathbf{1})$$

Standard Deviation

The square root of the variance, in the same units as $\mathbf{x}$.

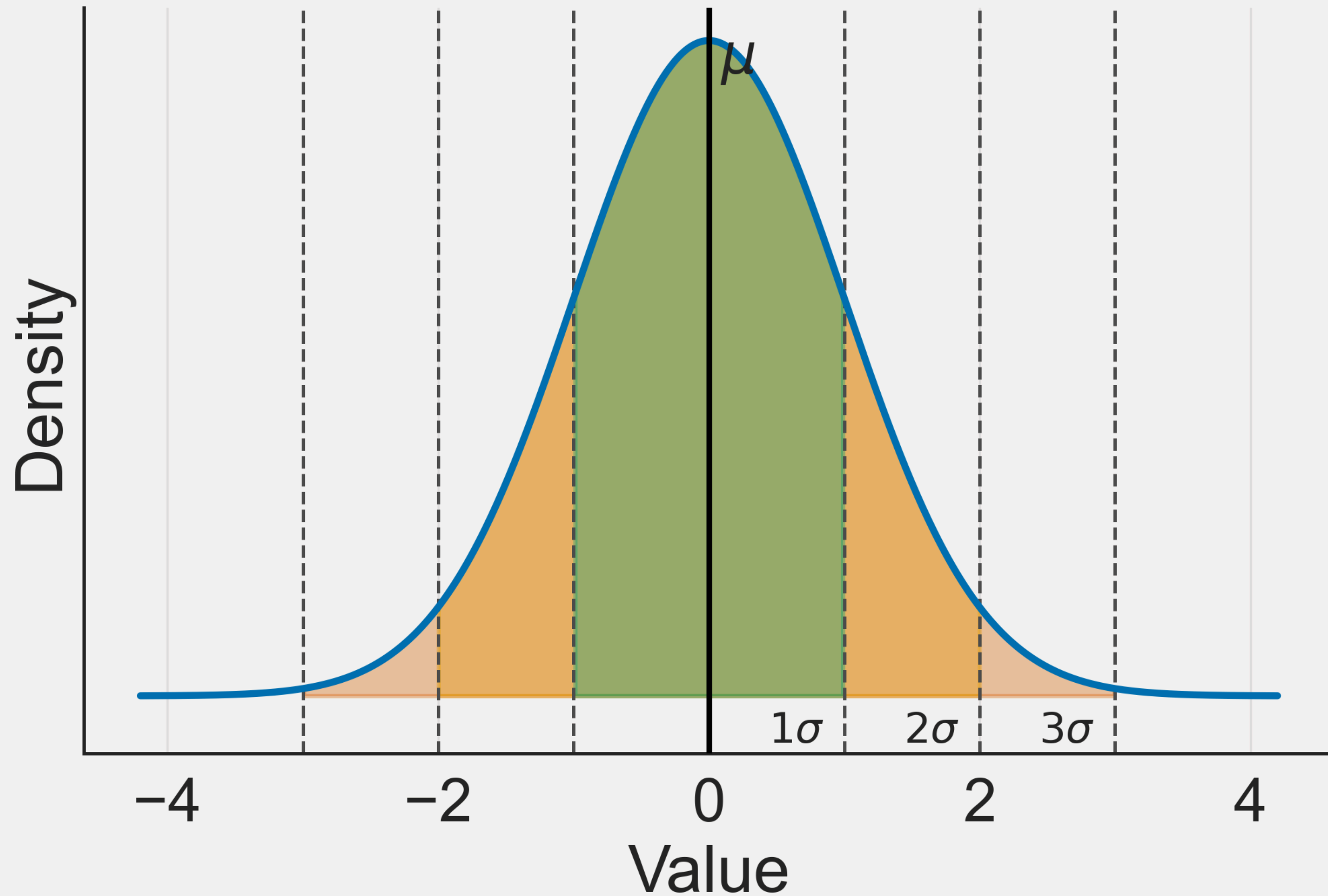
$$\sigma = \sqrt{\sigma^2} = \frac{1}{\sqrt{n}} \|\mathbf{x} - \mu \mathbf{1}\|_2$$

Sample version:

$$s = \sqrt{s^2} = \frac{1}{\sqrt{n-1}} \|\mathbf{x} - \bar{x} \mathbf{1}\|_2$$

Standard Deviation

Standard Deviation Marks Distance From the Mean



Z-Score

The number of standard deviations a value is from the mean.

$$Z_i = \frac{x_i - \mu}{\sigma}$$

In vector form:

$$\mathbf{z} = \frac{\mathbf{x} - \mu \mathbf{1}}{\sigma}$$

Notation

- σ^2 — population variance
- s^2 — sample variance
- σ, s — population and sample standard deviation, respectively
- $\text{Var}(X)$ — the variance operator, used in probability theory for the theoretical variance of a random variable X

Skewness

Skewness

The third moment. The asymmetry of a distribution around its mean.

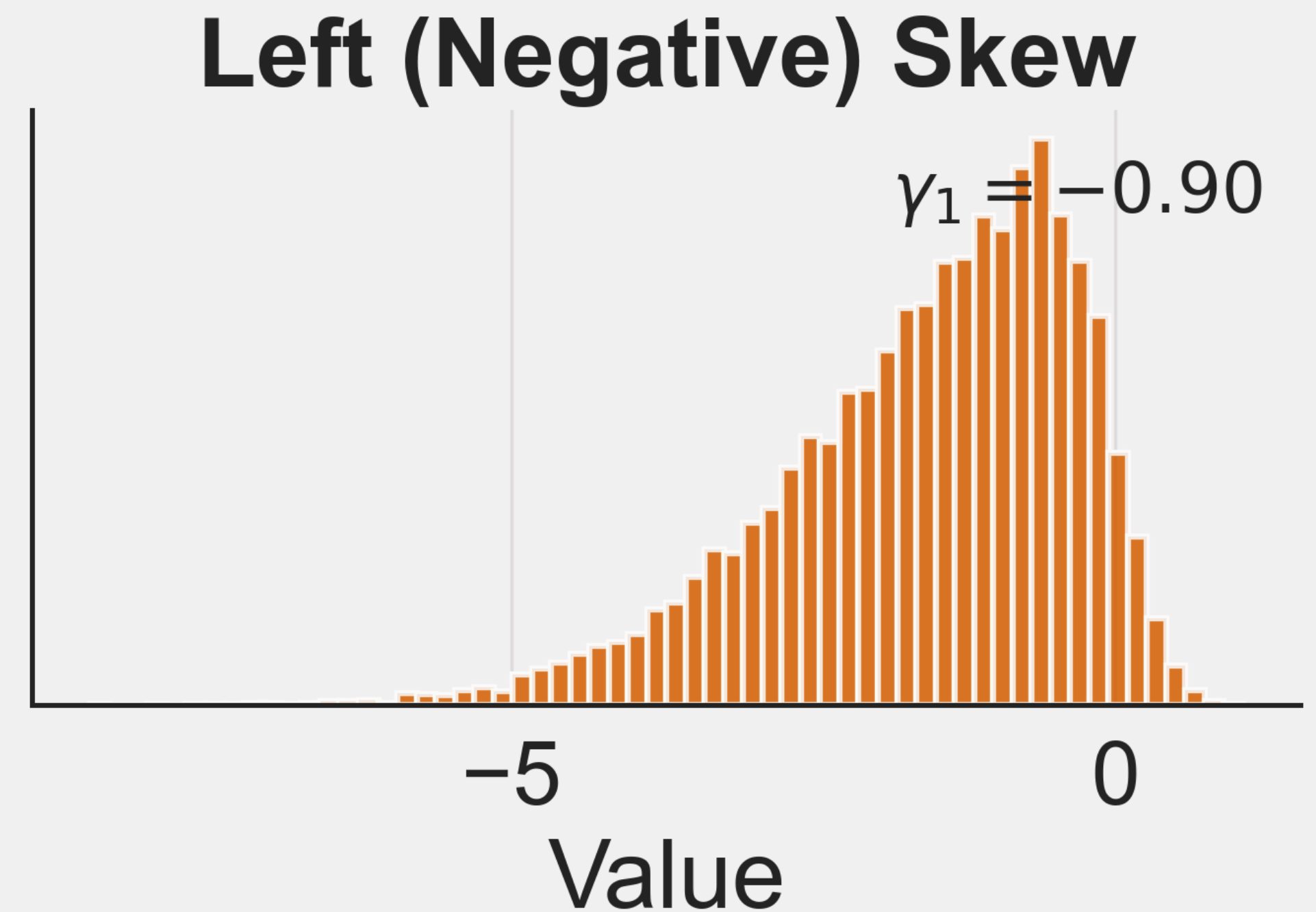
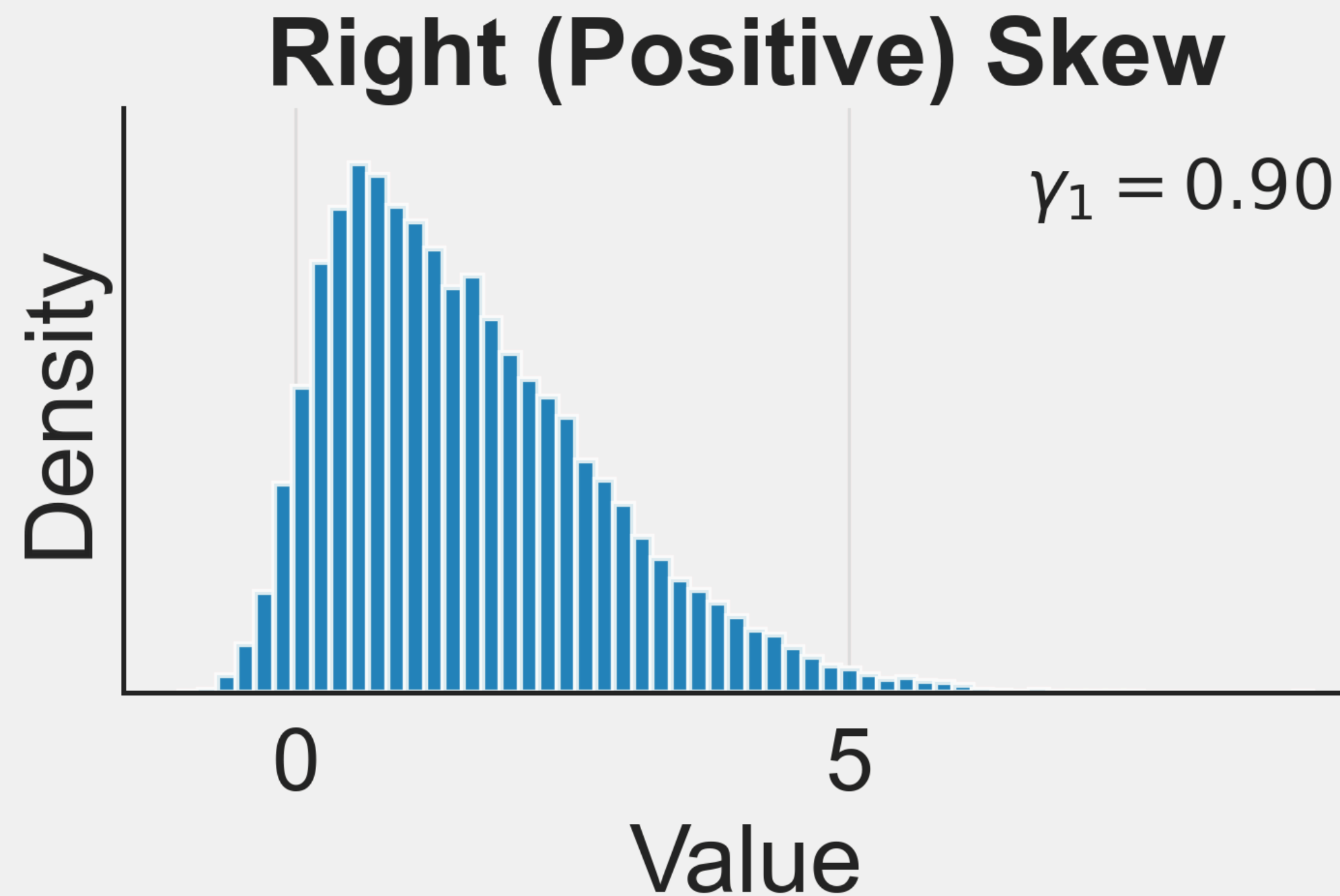
$$\gamma_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^3}{\sigma^3}$$

Sample version:

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{s^3}$$

Skewness

Skewness Captures the Direction of a Distribution's Tail



Notation

- γ_1 — population skewness
- g_1 — sample skewness

Covariance and Correlation

Covariance

A measure of how two random variables change together.

$$\sigma_{XY} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

Covariance is a dot product

Let $\tilde{\mathbf{x}} = \mathbf{x} - \mu_X \mathbf{1}$ and $\tilde{\mathbf{y}} = \mathbf{y} - \mu_Y \mathbf{1}$:

$$\sigma_{XY} = \frac{1}{n} \tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}$$

Pearson Correlation

Covariance rescaled to fall between -1 and 1.

$$\rho_{XY} = \frac{\sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)}{\sqrt{\sum_{i=1}^n (x_i - \mu_X)^2} \sqrt{\sum_{i=1}^n (y_i - \mu_Y)^2}}$$

As a dot product:

Let $\tilde{\mathbf{x}} = \mathbf{x} - \mu_X \mathbf{1}$ and $\tilde{\mathbf{y}} = \mathbf{y} - \mu_Y \mathbf{1}$:

$$\rho_{XY} = \frac{\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\|_2 \|\tilde{\mathbf{y}}\|_2}$$

Pearson Correlation

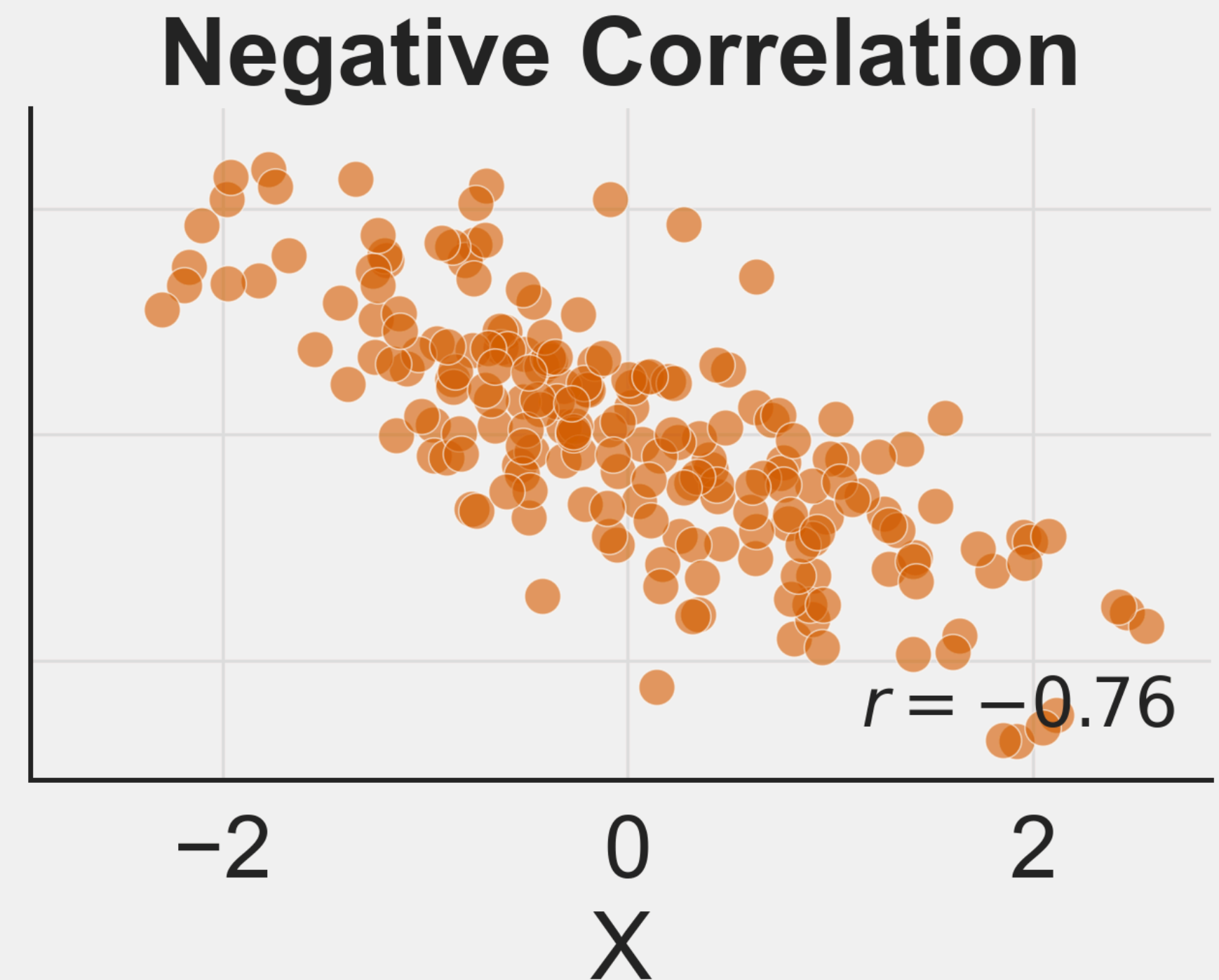
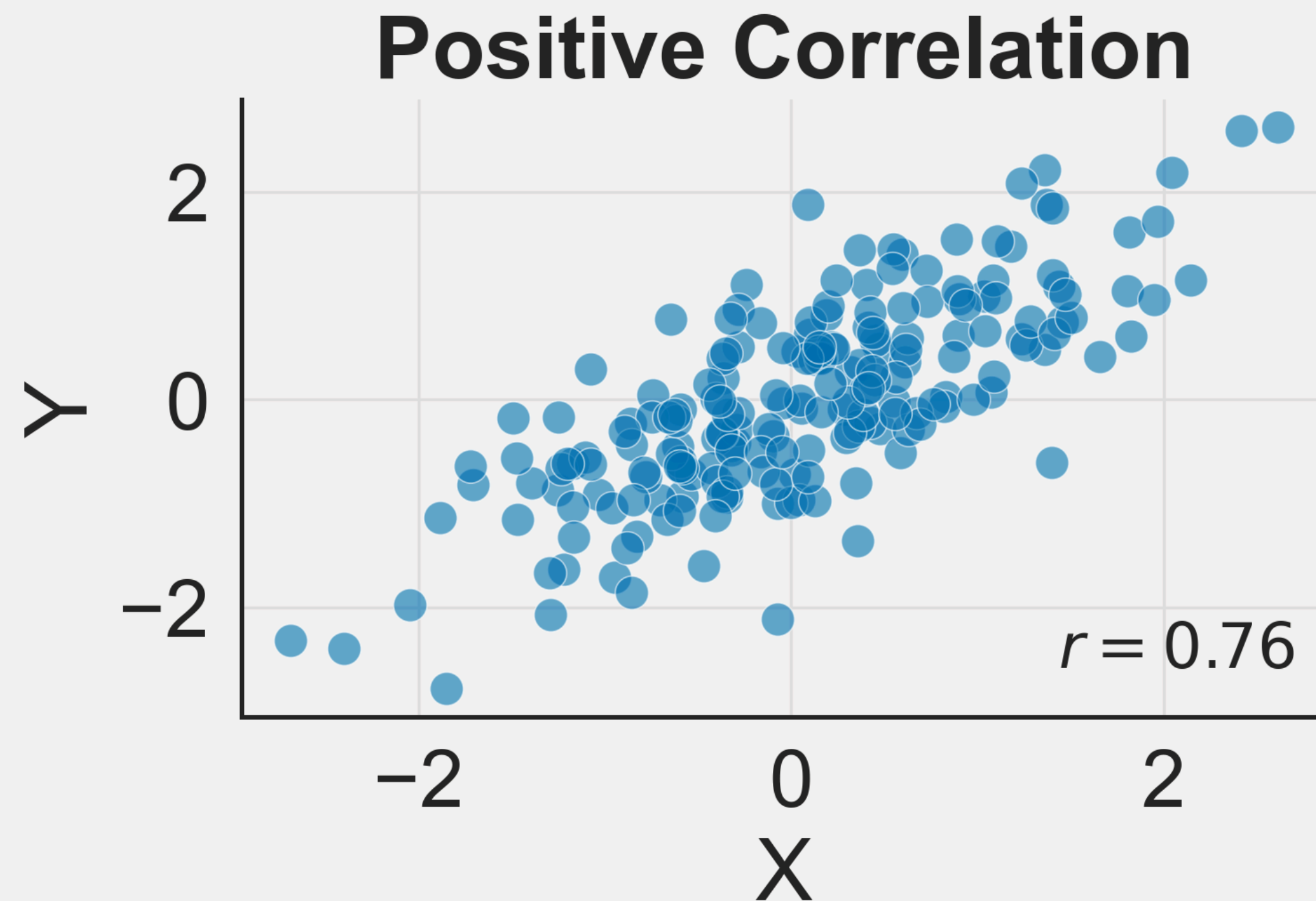
The correlation coefficient can also be expressed in terms of z-scores.

Let $\mathbf{z}_X = \frac{\tilde{\mathbf{x}}}{\sigma_X}$ and $\mathbf{z}_Y = \frac{\tilde{\mathbf{y}}}{\sigma_Y}$:

$$\rho_{XY} = \frac{1}{n} \mathbf{z}_X \cdot \mathbf{z}_Y$$

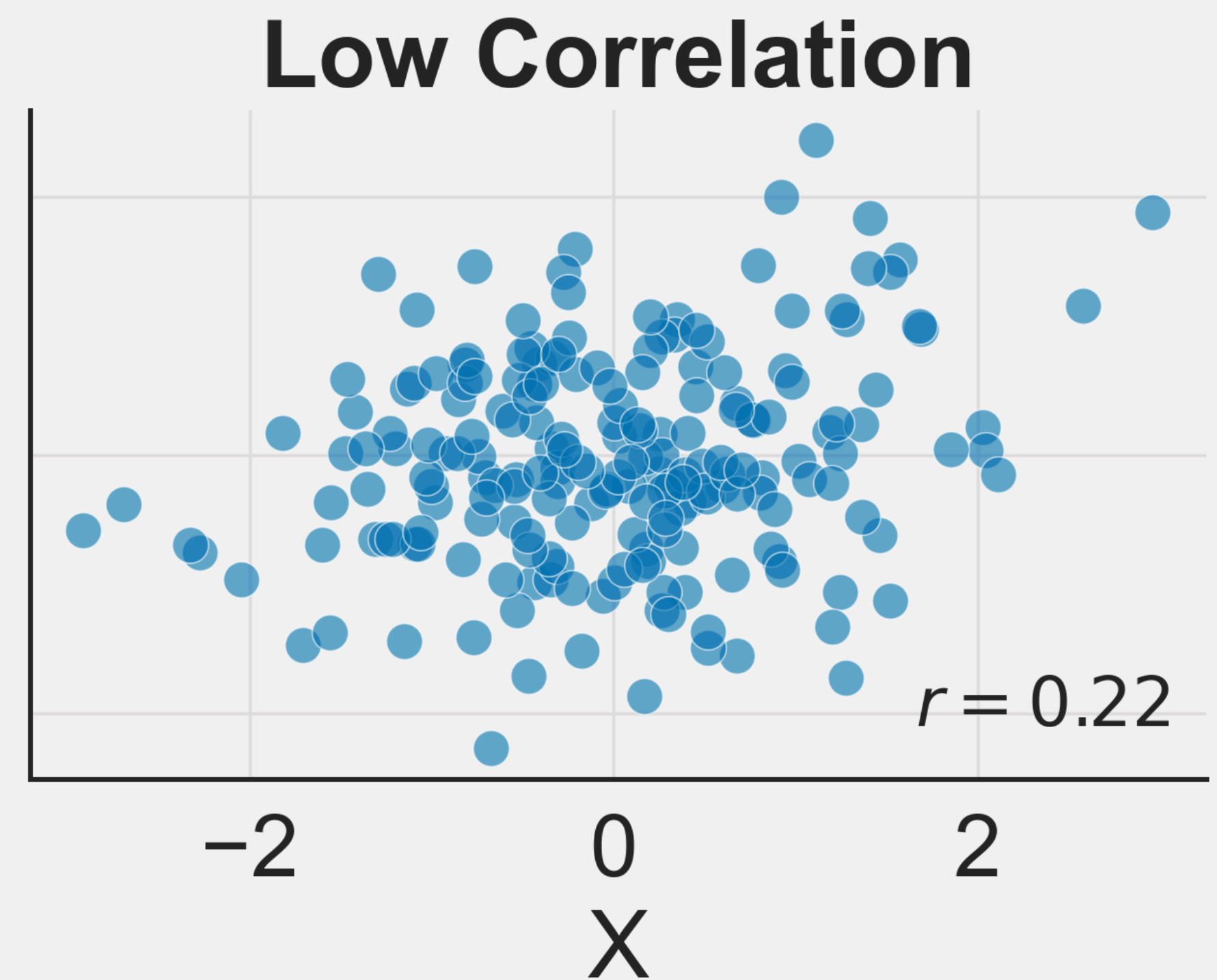
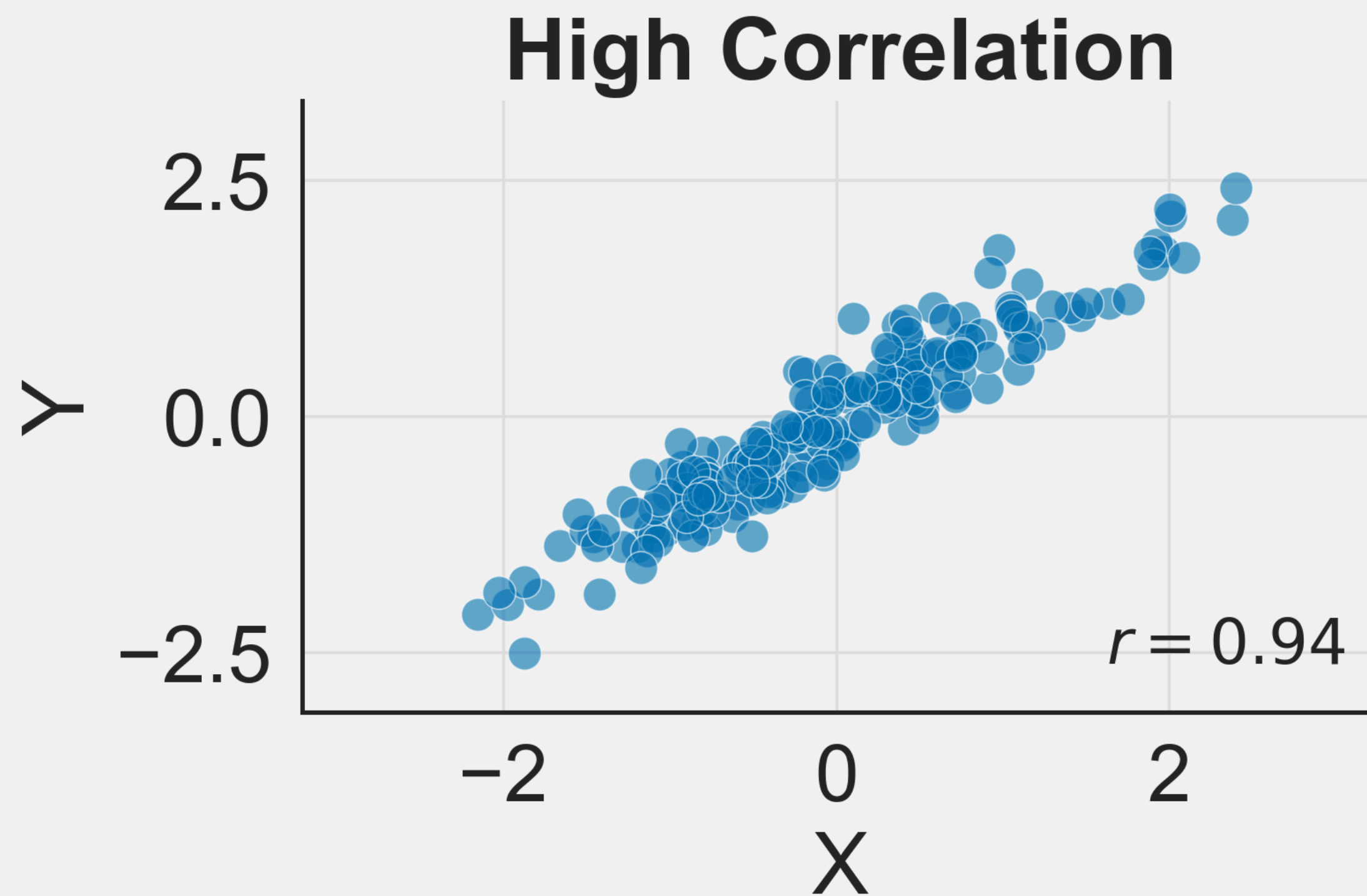
Correlation

Correlation Sign Reflects the Direction of the Relationship



Correlation

Correlation Strength Reflects How Tightly Points Cluster



Notation

- σ_{XY} — population covariance between X and Y
- $\text{Cov}(X, Y)$ - covariance operator
- s_{XY} — sample covariance
- ρ_{XY} — population correlation coefficient
- r_{XY} (often just r) — sample correlation coefficient (Pearson's r)

Vocabulary